

9.2.2

EE25BTECH11042 - Nipun Dasari

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Question

Find the points of intersection of the following parabola and circle and the region between them

$$y^2 = 6ax \quad (0.1)$$

$$x^2 + y^2 = 16a^2 \quad (0.2)$$

Solution: Standard Form and Matrices

First, we express the equations in the standard quadratic form $g(x, y) = 0$

$$y^2 - 6ax = 0 \quad (0.3)$$

$$x^2 + y^2 - 16a^2 = 0 \quad (0.4)$$

In homogeneous coordinates, the matrices for the conics are :

$$C_1 = \begin{pmatrix} 0 & 0 & -3a \\ 0 & 1 & 0 \\ -3a & 0 & 0 \end{pmatrix} \quad (0.5)$$

$$C_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -16a^2 \end{pmatrix} \quad (0.6)$$

Solution: The Degenerate Conic

The intersection points are on a conic formed by the linear combination $P_1 + \mu C_2 = 0$. We find μ by setting the determinant of the combined matrix $C(\mu) = C_1 + \mu C_2$ to zero.

The combined matrix is:

$$C(\mu) = \begin{pmatrix} \mu & 0 & -3a \\ 0 & 1 + \mu & 0 \\ -3a & 0 & -16a^2\mu \end{pmatrix}$$

By applying $R_3 \rightarrow R_3 + \frac{3a}{\mu}R_1$, we can find the determinant:

$$\begin{aligned} \det(C(\mu)) &= \det \begin{pmatrix} \mu & 0 & -3a \\ 0 & 1 + \mu & 0 \\ 0 & 0 & -16a^2\mu - \frac{9a^2}{\mu} \end{pmatrix} \\ &= (\mu)(1 + \mu) \left(-16a^2\mu - \frac{9a^2}{\mu} \right) \\ &= -a^2(1 + \mu)(16\mu^2 + 9) = 0 \end{aligned}$$

The only real root is $\mu = -1$.

Solution: Finding the Lines of Intersection

With $\mu = -1$, we subtract the circle's equation from the parabola's to get the degenerate conic :

$$(y^2 - 6ax) - (x^2 + y^2 - 16a^2) = 0$$

$$-x^2 - 6ax + 16a^2 = 0$$

$$x^2 + 6ax - 16a^2 = 0$$

Factoring the equation gives two vertical lines :

$$(x + 8a)(x - 2a) = 0 \tag{0.7}$$

The lines are $x = 2a$ and $x = -8a$. The line $x = -8a$ does not intersect the parabola $y^2 = 6ax$ in real points (for $a > 0$), so we use $x = 2a$.

Solution: Intersection Formula Setup

We find the intersection of the line $x - 2a = 0$ with the parabola P_1 using the formula $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$.

The parameters for the line and conic are :

$$\mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix}$$

$$\mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{u} = \begin{pmatrix} -3a \\ 0 \end{pmatrix}$$

$$f = 0$$

Solution: Calculating Parameter κ_i

First, we compute the necessary components :

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 1 \quad (0.8)$$

$$g(\mathbf{h}) = -12a^2 \quad (0.9)$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = 0 \quad (0.10)$$

Substituting these into the formula for κ_i :

$$\begin{aligned} \kappa_i &= \frac{1}{1} \left(-(0) \pm \sqrt{(0)^2 - (-12a^2)(1)} \right) \\ &= \pm \sqrt{12a^2} = \pm 2\sqrt{3}a \end{aligned} \quad (0.11)$$

Solution: Final Intersection Points

Finally, we find the points using $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$:

For $\kappa_1 = 2\sqrt{3}a$:

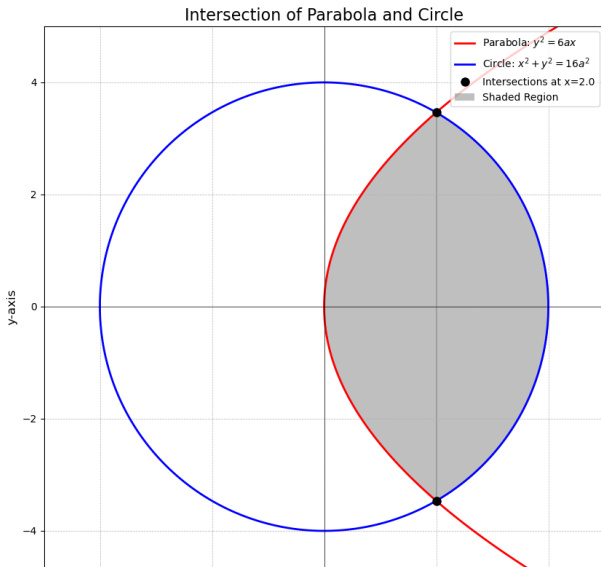
$$\mathbf{x}_1 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} + 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix} \quad (0.12)$$

For $\kappa_2 = -2\sqrt{3}a$:

$$\mathbf{x}_2 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} - 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \quad (0.13)$$

The intersection points are $(2a, 2\sqrt{3}a)$ and $(2a, -2\sqrt{3}a)$.

Plot: Parabola and Circle



Plot: Intersection Points

